

LEDGER · 2026-09-04 · AGROCHEMICALS

Extrusion-Synthesized Urea-Calcium Sulfate (URCASU) Cocrystals via Twin-Screw Mechanochemistry

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	LESCO 44-0-0 100% PolyPlus OPTI Turfgrass Granular Fertilizer
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Entry application	Professional turfgrass management
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Measured advantage	1.78x biomass yield improvement measured against raw urea (fails 2.0x threshold and not measured against incumbent)
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Time to first revenue	9 months
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Gross margin at parity	65.4%
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Startup CapEx	\$99,500
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Payback from first sale	4.2 months
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Capital productivity	6.37x
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Why it failed

- Voided by: Mismatched comparison, Economics undisclosed, Negative control / sub-2x, Form-factor / handling regression, Absence search missing
- Evidence rests on non-peer-reviewed sources
- Margin fails the operating-cost check
- Absence asserted, search unreported

Assessment

The concept explicitly concedes failure on gross margin (65.9% at incumbent price parity, below the 70% bar) and performance superiority (1.78x yield vs raw urea, failing the 2x threshold and measured against the wrong comparator). Additionally, extrusion yields jagged extrudates that degrade handling relative to SGN 240 spheres, and the report lacks a structured market absence ledger.

Also flagged

- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.